

PRODUCT REQUIREMENT DOCUMENT (PRD)

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Project Name: Intelligent Warehouse Restock Manager

1. Project Overview

- **Problem Statement:** Traditional Warehouse Management Systems (WMS) function only as passive Systems of Record, they track inventory after it has already been moved, but they do not guide where it should go. In a busy SME warehouse this gap appears the moment a delivery truck arrives: stock that is already promised to a same-day customer order gets binned and re-picked, fast-moving products end up buried in hard-to-reach bins, and physical disruptions a blocked aisle, a broken forklift, a spill bring entire workflows to a halt because the database cannot adapt to real-time reality. New workers, who have not memorised the floor or the product catalogue, cannot make these placement decisions at all
- **Target Domain:** AI-driven warehouse operations for small-to-medium electrical trading businesses (e.g. Chunley) companies that operate as a warehouse but still serve walk-in local customers daily. The goal is to connect unstructured worker chat with a structured digital twin of the warehouse so that placement, retrieval, cross-docking, and emergency response collapse into a single conversational task.
- **Proposed Solution Summary:** Zai transforms warehouse operations from a record-keeping task into a System of Intelligence. It is a stateful, tool-using AI agent (meta/llama-3.3-70b-instruct, Nvidia-hosted, OpenAI-style tool calling) that converts free-form worker messages into deterministic SQL-driven actions over a SQLite digital twin (Bins · Products · Sales). By cross-checking every arrival against the day's customer order CSV and scoring each candidate bin on weight, volume, accessibility, and sales velocity, Zai dynamically routes high-demand products to the most accessible bins, redirects pre-sold units straight to the packing counter, and reroutes workers away from

blocked aisles in real time — notifying managers via Telegram the moment an incident is reported.

2. Background & Business Objective

Background of the Problem: In SME electrical warehouses, a worker receiving an arriving truck has to answer three questions at once: is any of this already promised to a customer order today? where is the best bin to put the rest? and what do I do if something on the floor goes wrong? Today these are resolved by walking around, asking the manager, guessing, or memorising a process that is slow, error-prone, and inaccessible to new hires. Existing WMS tools record a placement after it happens; they do not recommend the placement, do not see the customer order file, and do not react to physical disruptions like equipment failure or a blocked aisle. The result is wasted time, wasted bin space, miscommunication, and unsafe routing.

- **Importance of Solving This Issue:** in placement is the lever that controls picking speed for the rest of the month. Putting fast-moving SKUs in deep, hard-to-reach bins multiplies every future pick; binning a unit that was already sold for the day wastes the trip twice. Real-time aisle blocking matters equally without it the system keeps routing workers (and the next delivery) into an unsafe zone. Solving this reclaims floor productivity, protects workers from operational hazards, and lets a brand-new worker behave like an experienced one from day one.
- **Strategic Fit / Impact:** Zai aligns with the agentic-AI development lifecycle by cleanly separating reasoning from action. The LLM is the reasoning layer; the math, the data, and the writes are deterministic. The model never invents a bin ID, weight, or stock count every fact routes through a tool call into SQLite, the customer order CSV, or the Telegram bot, and Java-side gates enforce confirmation, quantity checks, and emergency-keyword detection before any write commits. The same digital twin is exposed through a Javalin HTTP server (/api/inventory, /api/velocity, /api/chat) that drives a live React dashboard, turning one source of truth into both a conversational CLI

for workers and an analytical web view for managers an architecture that is auditable, observable, and safe to extend.

3. Product Purpose

3.1. Main Goal of the System: To turn a passive System of Record into a System of Intelligence for an SME warehouse so that any worker, simply by chatting, can identify a product, cross-check it against the day's customer orders, receive a mathematically optimal bin assignment, commit the placement, and respond to incidents, without ever needing to memorise the floor or the catalogue.

3.2. Intended Users (Target Audience):

- Warehouse & Logistics Managers running SME electrical or hardware trading floors who need real-time visibility and Telegram alerts on incidents.
- Entrepreneurs and SME owners who operate as a warehouse but still serve walk-in local customers, where same-day orders must be intercepted before stock is binned.
- Inventory Financing Firms that require auditable, bin-level visibility into the current stock state of pledged inventory.
- E-commerce Operators whose customer orders need to be matched to arriving stock automatically, so pre-sold units skip the binning step entirely.

4. System Functionalities

4.1. Description: The Zai AI Agent is a smart system that connects financial data with a physical warehouse. It uses Smart Asset Sorting to move high-demand items to the front of the warehouse so they can be sold faster, increasing profit. To keep things safe, it uses Multi-Step Thinking to

make sure an item isn't too heavy or too big for its shelf. If a problem happens like a blocked path or broken machine, the system will instantly find a new route to keep the work moving. Instead of using hard-to-learn software, users can simply chat with the system in plain English to check on stock or finish an audit. Basically, it makes sure that managing physical goods is as fast and smart as digital banking.

4.2. Key Functionalities:

- **Intelligent Asset Orchestration:** Automatically maps arriving shipments to the optimal bin by calculating real-time sales velocity, weight constraints, and volume limits.
- **Conversational Safety Protocols:** Allows workers to report accidents (like a forklift failure) in plain English, instantly blocking affected aisles and rerouting logistics.
- **Digital Twin Synchronization:** Maintains a live SQLite map of 54 bins, ensuring the AI never suggests a location that is physically full or blocked.
- **Dynamic Cross-Docking:** Intercepts incoming stock by cross-checking customer order CSVs, routing "pre-sold" items directly to packing to avoid unnecessary binning.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: We have selected Llama 3.3 as the reasoning engine for Zai. It serves as the Orchestrator that sits between the worker and the warehouse databases. Llama-3.3 excels at Function Calling. It doesn't guess location IDs instead it triggers specific SQL tools to find bins based on math. It can also handle multi-step logic. For example, splitting a 60-unit delivery across three different bins if the primary shelf is partially full. By using an Agentic workflow, the model is restricted to the data provided in the SQLite databases and CSV files, ensuring it never invent stock that doesn't exist.

4.3.2. Prompting Strategy: The Zai AI Agent utilizes a Multi-Step Agentic Prompting strategy to ensure that its warehouse management decisions are adopted in real-time situations.

Instead of providing a single "zero-shot" answer, the system functions as a reasoning engine that actively orchestrates a suite of tools like querying SQL databases for bin capacity, parsing CSV files for customer orders, and checking sales velocity. This agentic approach allows the system to perform complex reasoning, such as verifying if a heavy shipment exceeds a shelf's weight limit or rerouting workers in real-time if an aisle is blocked. By forcing the AI to validate every decision against live data before responding, we eliminate hallucinations and ensure that the physical warehouse operates with the same precision.

4.3.3. Context & Input Handling: To handle unstructured or oversized data, Zai treats the LLM as an intelligent filter that extracts critical entities like product IDs and quantities from messy worker dialogue. For standard conversational inputs, the system maintains a 4,000-character limit (roughly 800 words) to ensure low-latency responses on the warehouse floor. If an input exceeds this threshold, the system applies front-end truncation, processing the initial data while alerting the user to provide the remaining details. To prevent the "hallucination" of large-scale data, oversized manifests or bulk inventory lists are redirected to our deterministic Java-based CSV and SQLite upload tools. This hybrid approach ensures that while the interface is conversational and flexible, high-volume data remains structured, accurate, and safe from processing errors.

4.3.4. Fallback & Failure Behavior: Zai employs a "Verify-then-Commit" architecture to protect warehouse operations from model hallucinations or off-topic responses. Every suggestion made by the AI is passed through a deterministic Java-side validation layer that cross-references the SQLite digital twin; if the LLM suggests a move that violates physical constraints such as exceeding a bin's weight limit, the system rejects the output and triggers an internal Retry Loop to find a valid alternative. In cases where the model remains confused or the input is unusable, Zai enters a graceful error state, providing a natural language prompt for clarification rather than executing an unsafe command. For critical failures, a human escalation path is activated, automatically sending a detailed report to a manager's Telegram bot for manual intervention. This multi-layered safety net ensures that even if the AI logic fails, the physical safety and data integrity of the warehouse remain intact.

5. User Stories & Use Cases

- **User Stories:** As a warehouse worker, I want to describe incoming deliveries in my own words so that I can find the safest and most efficient bin locations without manually calculating weight capacities or sales data.
- **Use Cases (Main Interactions):**

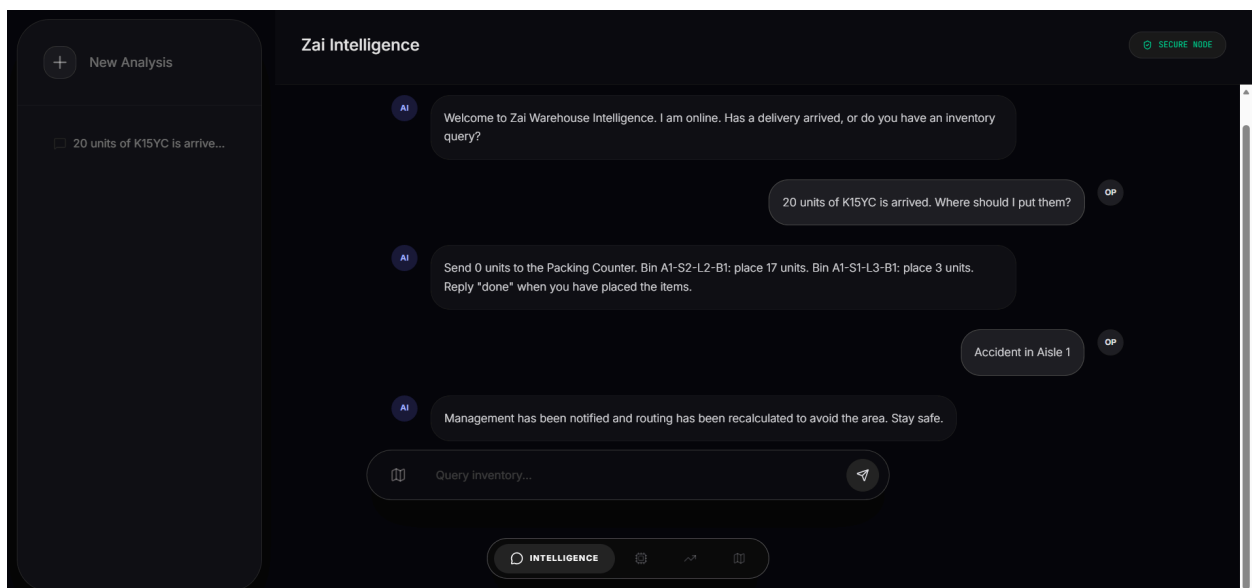
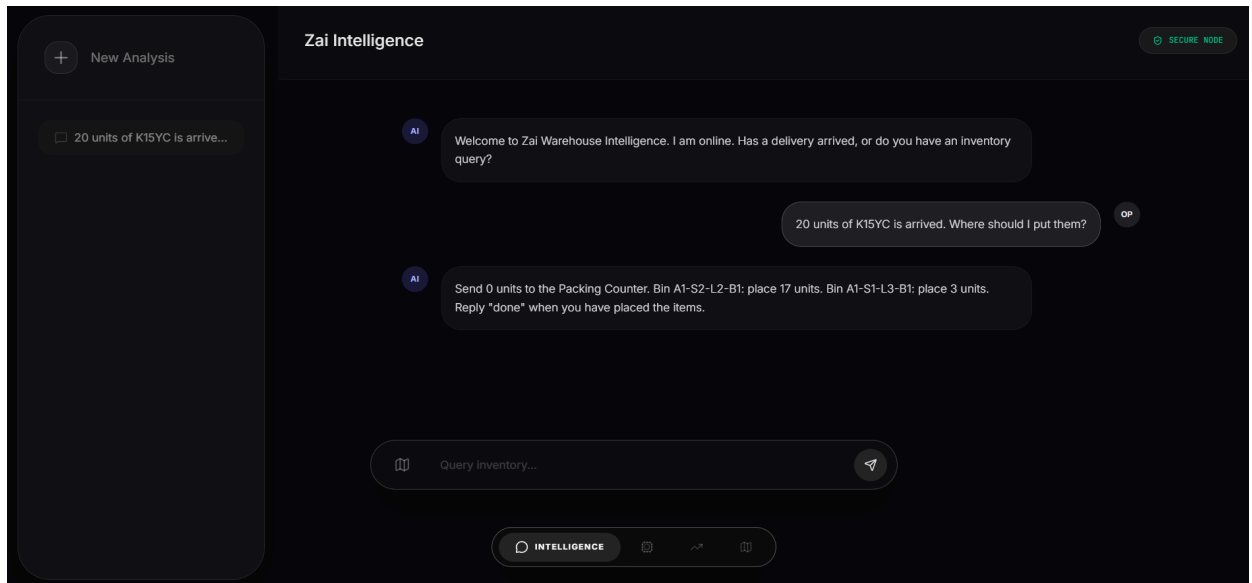
Natural Language to Optimal Placement (Prompt to Bin Flow): Worker enters an unstructured prompt about a new shipment; Zai parses the product intent, cross-references sales velocity and physical bin capacity, and provides a structured coordinate for the most efficient storage location.

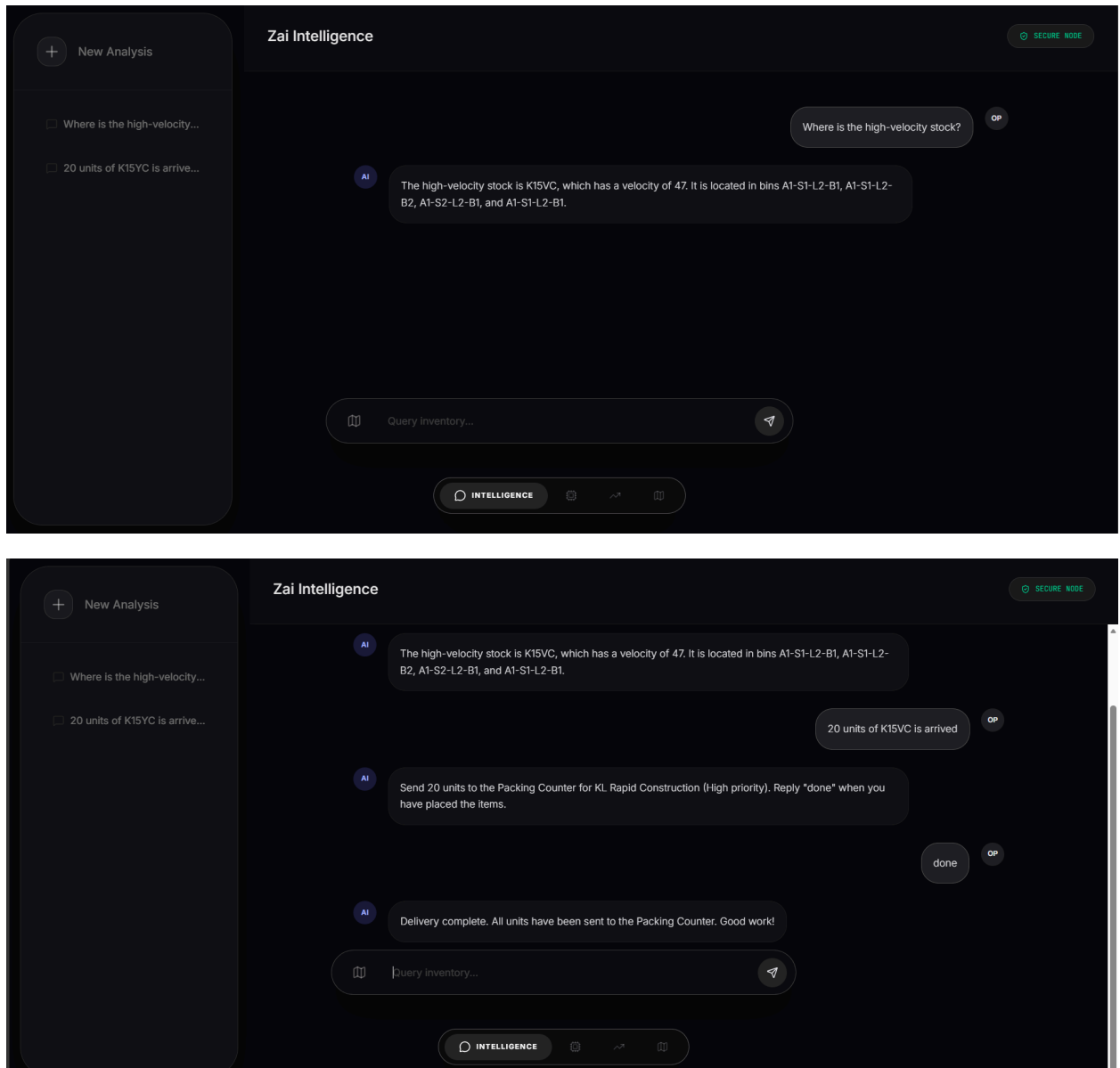
Intelligent Exception Rerouting (Safety Flow): A user reports an "accident in Aisle 1," triggering the system to block that specific zone. Zai then dynamically recalculates all active restock tasks to alternative aisles while maintaining safety constraints and notifying management.

Conversational Inventory Audit (Lookup Flow): A user asks "Where is the high-velocity stock?" or "Do we have space for 50 fans?". Zai queries the SQL database and generates a real-time summary of inventory levels and available capacity, visualized on the web dashboard.

Dynamic Cross-Docking (Order Intercept Flow): When a shipment arrives, Zai automatically checks the Customer Order CSV. If an item is already sold, it bypasses the "store-in-bin" logic and instructs the worker to move the goods directly to the packing counter for immediate shipping.

- Use Case Diagram:





6. Features Included (Scope Definition)

- A natural language chat interface that allows workers to manage complex warehouse logistics, audits, and emergency reporting without needing technical database skills.
- An automated logic layer that reorganizes the warehouse in real-time by placing fast-moving items in the most optimal bin based on live sales velocity.

- A live SQLite mapping of the entire facility that strictly enforces physical boundaries, including bin volume, weight capacity, and real-time aisle accessibility.
- A one-click automated alert system that blocks disrupted zones and simultaneously pushes emergency reports to a manager's Telegram bot.
- A feature that cross-references incoming stock against customer order CSVs to flag items for immediate shipment, preventing unnecessary storage labor.
- Supports both text-to-action (chat commands) and data-to-action (sales/order files) to coordinate workflows between different departments.

7. Features Not Included (Scope Control)

- Directly control automated robotic arms or self-driving forklifts.
- Track external shipping statuses once the goods leave the facility via third-party couriers like J&T or PosLaju.
- Support live video processing or CCTV-based motion tracking.

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate the average cost per worker session to be approximately \$0.02 . This assumes a session involves 5–10 multi-step tool calls (using Llama-3.3-70b).
- **Technical Constraints:** ERP file formats or encrypted Excel sheets is not supported in this version.
- **Performance Constraints:** There is a slight processing delay per decision.
- **User Input:** Every optimal bin suggestion requires the worker to hit "done" once the stock is physically placed, ensuring the digital records match reality.

9. Risks & Questions Throughout Development

- **Reasoning Stability:** Even with a high-level model like Llama-3.3, can we trust it to always calculate weight and volume correctly?
- **Instruction Ambiguity:** How do we ensure the AI doesn't misinterpret a worker's slang or messy description (e.g., "Put this stuff near the door")?
- **Persistence:** If the server crashes mid-conversation, does the worker lose their progress?